Chapter 6-7: Convex Sets and Functions

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(This slide is mainly modified from slides of S. Boyd EE364a, Stanford, and R. Tibshirani 10-725, CMU.)

Convex optimization problem

Consider the general optimization problem

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_j(x) = 0, \quad j = 1, \ldots, p \end{array}$$

Its domain is the common domain of all the functions, i.e.

$$\mathcal{D} = \mathrm{dom} f \cap \bigcap_{i=1}^{m} \mathrm{dom} f_i \cap \bigcap_{j=1}^{p} \mathrm{dom} h_j.$$

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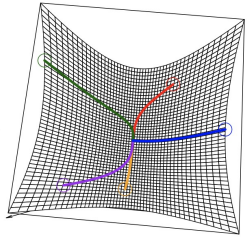
$$\mathcal{D} = \mathrm{dom} f \cap \bigcap_{i=1}^{m} \mathrm{dom} f_i \cap \bigcap_{j=1}^{p} \mathrm{dom} h_j.$$

This is a convex optimization problem provided the function f_0, f_i, $i = 1, \ldots, m$ are convex, and h_j, $j = 1, \ldots, p$ are affine, i.e. the equality constraints can be only $Ax = b$.

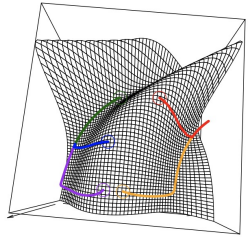
Local minima are global minima

For convex optimization problems, local minima are global minima.

This is a very useful fact and will save us a lot of trouble!



Convex



Nonconvex

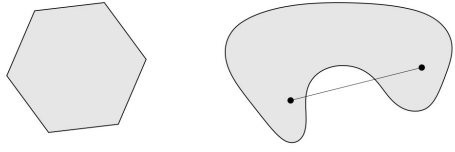
Convex Sets

Convex Sets

Convex set: $C \subseteq \mathbb{R}^n$ such that

$$x, y \in C \quad \Longrightarrow \quad tx + (1-t)y \in C \text{ for all } 0 \leq t \leq 1$$

In words, line segment joining any two elements lies entirely in set.



Convex combination of $x_1, \ldots, x_k \in \mathbb{R}^n$: any linear combination $\theta_1 x_1 + \cdots \theta_k x_k$ with $\theta_i \geq 0, i = 1, \ldots, k$ and $\sum_{i=1}^k \theta_i = 1$.

Convex hull of a set C, $\mathrm{conv}(C)$, is all convex combinations of elements. Always convex.

Examples of convex sets

- Trivial ones: empty set, point, line, $\mathbb{R}^n$
- Affine space: $\{x : Ax = b\}$, for given A, b
- Norm ball: $\{x : \|x\| \leq r\}$, for given norm $\|\cdot\|$ and radius r
- Hyperplane: $\{x : a^T x = b\}$, for given a, b
- Halfspace: $\{x : a^T x \leq b\}$
- Polyhedron: $\{x : Ax \preceq b, Cx = d\}$, where inequality $\preceq$ is the componentwise.

Examples of convex sets (cont.)

- Norm ball: $\{x : \|x\| \leq r\}$, for given norm $\|\cdot\|$ and radius r

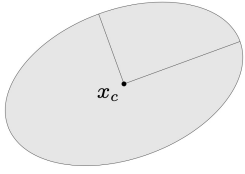
Example: **(Euclidean) ball** with center x_c and radius r:

$$B(x_c, r) = \{x : \|x - x_c\|_2 \leq r\}$$

Example: **ellipsoid**: set of the form

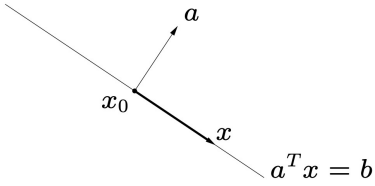
$$\{x : \|x - x_c\|_{P^{-1}} \leq 1\} = \{x : (x - x_c)^T P^{-1}(x - x_c) \leq 1\}$$

with $P \in \mathbb{S}^n_{++}$ (i.e. symmetric positive definite matrices)

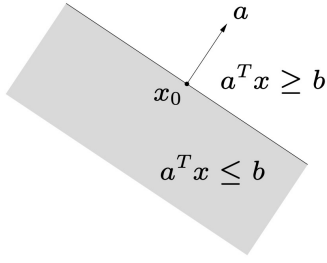


Examples of convex sets (cont.)

- Hyperplane: $\{x : a^T x = b\}$, $(a \neq 0)$



- Halfspace: $\{x : a^T x \leq b\}$, $(a \neq 0)$

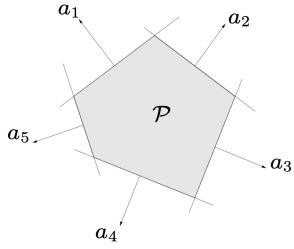


Examples of convex sets (cont.)

- Polyhedron: solution set of finitely many linear inequalities and equalities

$$\{x : Ax \preceq b, Cx = d\}$$

where $A \in \mathbb{R}^{m\times n}, C \in \mathbb{R}^{p\times n}$, $\preceq$ is componentwise inequality.



polyhedron is intersection of finite number of halfspaces and hyperplanes.

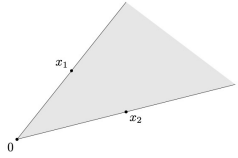
Cones

Cone: $C \subseteq \mathbb{R}^n$ such that

$$x \in C \quad \Longrightarrow \quad tx \in C \text{ for all } t \geq 0$$

Convex cone: cone that is also convex, i.e.

$$x_1, x_2 \in C \quad \Longrightarrow \quad t_1 x_1 + t_2 x_2 \in C \text{ for all } t_1, t_2 \geq 0$$

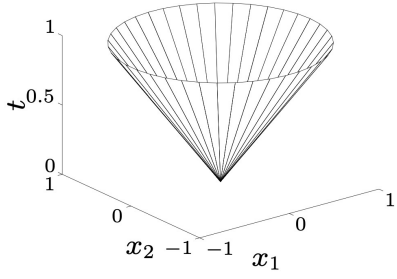


Conic combination of $x_1, \ldots, x_k \in \mathbb{R}^n$: any linear combination $\theta_1 x_1 + \cdots + \theta_k x_k$ with $\theta_i \geq 0, i = 1, \ldots, k$. Conic hull collects all conic combinations.

Examples of convex cones

- Norm cone: $\{(x, t) : \|x\| \leq t\}$, for a norm $\|\cdot\|$.

  Under the ℓ_2 norm $\|\cdot\|_2$, is called second-order cone.



Examples of convex cones

- Norm cone: $\{(x, t) : \|x\| \leq t\}$, for a norm $\|\cdot\|$.

  Under the ℓ_2 norm $\|\cdot\|_2$, is called second-order cone.

- Positive semidefinite cone:

$$\mathbb{S}^n_+ = \{X \in \mathbb{S}^n : X \succeq 0\},$$

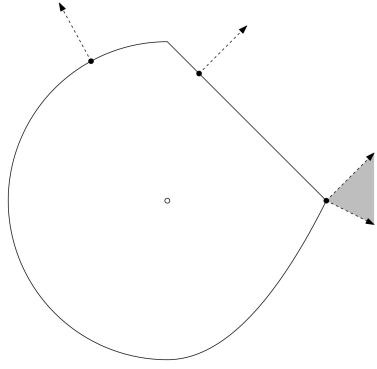
  where $X \succeq 0$ means X is positive semidefinite, $\mathbb{S}^n$ is the set of $n \times n$ symmetric matrices.

  (Additional notation: $\mathbb{S}^n_{++} = \{X \in \mathbb{S}^n : X \succ 0\}$: positive definite $n \times n$ matrices)

Examples of convex cones

- Normal cone: given any set C and point $x \in C$,

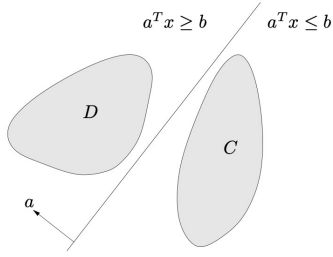
$$\mathcal{N}_C(x) = \{g : g^T x \geq g^T y, \text{ for all } y \in C\}.$$



This is always a convex cone, regardless of C

Key properties of convex sets

Separating hyperplane theorem: two disjoint convex sets have a separating hyperplane between them

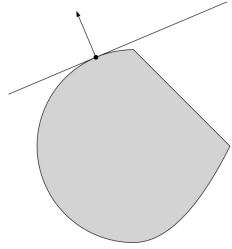


Formally: if C, D are nonempty convex sets with $C \cap D = \emptyset$, then there exists a, b such that

$$C \subseteq \{x : a^T x \leq b\}, \qquad D \subseteq \{x : a^T x \geq b\}$$

Key properties of convex sets

Supporting hyperplane theorem: a boundary point of a convex set has a supporting hyperplane passing through it



Formally: if C is a nonempty convex set, and $x_0 \in \partial C$, then there exists a such that

$$C \subseteq \{x : a^T x \leq a^T x_0\}$$

Operations preserving convexity

- Intersection: the intersection of convex sets is convex
- Scaling and translation: if C is convex, then

$$aC + b = \{ax + b : x \in C\}$$

  is convex for any a, b
- Affine images and preimages: if $f(x) = Ax + b$ and C is convex then

$$f(C) = \{f(x) : x \in C\}$$

  is convex; and if D is convex then

$$f^{-1}(D) = \{x : f(x) \in D\}$$

  is convex.

Example: solution set of linear matrix inequality

Given $A_1, \ldots, A_m \in \mathbb{S}^n$, the solution set C of linear matrix inequality is

$$C \coloneqq \{x : x_1 A_1 + x_2 A_2 + \cdots + x_m A_m \preceq B\}$$

The set C is convex.

Solution: let $f : \mathbb{R}^m \to \mathbb{S}^n$, $f(x) = B - \sum_{i=1}^n x_i A_i$. Then

$$C = f^{-1}(\mathbb{S}^n_+)$$

the affine preimage of convex set.

Operations preserving convexity (cont.)

- Perspective function $P : \mathbb{R}^{n+1} \to \mathbb{R}^n$:

$$P(x,t) = x/t, \quad \mathrm{dom} P = \{(x,t) : t > 0\}$$

  Images and preimages of convex sets under perspective are convex.

- Linear-fractional function $f : \mathbb{R}^n \to \mathbb{R}^m$:

$$f(x) = \frac{Ax + b}{c^T x + d}, \quad \mathrm{dom} f = \{x : c^T x + d > 0\}$$

  Images and preimages of convex sets under linear-fractional functions are convex.

Convex Functions

Convex functions

Convex function: $f : \mathbb{R}^n \to \mathbb{R}$ such that $\mathrm{dom}(f) \subseteq \mathbb{R}^n$ is convex and

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y), \quad t \in [0,1]$$

for all $x, y \in \mathrm{dom} f$.



That is, function lies below the line segment joining $f(x), f(y)$.

Concave function: f concave $\iff -f$ convex

Convex functions

Important modifers:

- strictly convex: for $x \neq y$ and $0 < t < 1$,

$$f(tx + (1-t)y) < tf(x) + (1-t)f(y).$$

  That is, f is convex and has greater curvature than a line function.

- strongly convex with parameter $m > 0$: $f - \frac{m}{2}\|x\|_2^2$ is convex.

  That is, f is at least as convex as a quadratic function.

Note: strongly convex $\Rightarrow$ strictly convex $\Rightarrow$ convex

Examples of convex functions

- Univariate functions:
  - exponential function: e^{ax} is convex for any a over $\mathbb{R}$
  - power function: x^a is convex for $a \geq 1$ or $a \leq 0$ over $\mathbb{R}_+$ (nonnegative reals)
  - power function: x^a is concave for $0 \leq a \leq 1$ over $\mathbb{R}_+$
  - logarithmic function: $\log x$ is concave over $\mathbb{R}_{++}$
- Affine function: $a^T x + b$ is both convex and concave
- Quadratic function: $\frac{1}{2} x^T P x + q^T x + c$ is convex provided that $P \succeq 0$ (positive semidefinite)
- Least squares loss: $\|y - Ax\|_2^2$ is always convex (why? since $A^T A$ is always positive semidefinite)

Examples of convex functions (cont.)

- Norm: $\|x\|$ is convex for any norm; e.g., ℓ_p norms,

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p} \text{ for } p \geq 1; \qquad \|x\|_\infty = \max_{i=1,\ldots,n} |x_i|$$

and also matrix operator (spectral) and trace (nuclear) norms,

$$\|X\|_{\text{op}} = \sigma_1(X), \qquad \|X\|_{\text{tr}} = \sum_{i=1}^{r} \sigma_r(X)$$

where $\sigma_1 \geq \cdots \geq \sigma_r \geq 0$ are the singular values of matrix X.

Examples of convex functions (cont.)

- Indicator function if C is convex, then its indicator function $I_C(x)$ is convex

$$I_C(x) = \begin{cases} 0 & x \in C \\ \infty & x \notin C \end{cases}$$

- Support function: for any set C (convex or not), its support function $I_C^*(x)$ is convex

$$I_C^*(x) = \max_{y \in C} x^T y$$

- Max function: $f(x) = \max\{x_1, \ldots, x_n\}$ is convex

Key properties of convex functions

Restriction of a convex function to a line: $f : \mathbb{R}^n \to \mathbb{R}$ is convex if and only if the function $g : \mathbb{R} \to \mathbb{R}$,

$$g(t) = f(x + tv), \quad \mathrm{dom} g = \{t : x + tv \in \mathrm{dom} f\}$$

is convex (in t) for any $x \in \mathrm{dom} f, v \in \mathbb{R}^n$.

Example: $f : \mathbb{S}^n \to \mathbb{R}$ with $f(X) = \mathrm{logdet} X$, $\mathrm{dom} f = \mathbb{S}^n_{++}$

$$g(t) = \mathrm{logdet}(X + tV) = \mathrm{logdet} X + \mathrm{logdet}(I + tX^{-1/2} V X^{-1/2}) = \mathrm{logdet} X + \sum_{i=1}^{n} \log(1 + t\lambda_i)$$

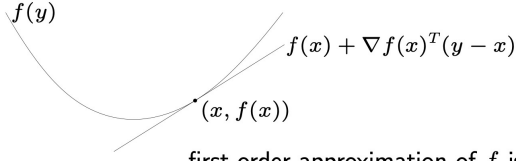
where λ_i are the eigenvalues of $X^{-1/2} V X^{-1/2}$. g is concave in t (for any choice of $X \succ 0, V$), hence f is concave.

Key properties of convex functions (cont.)

First-order condition:

If f is differentiable, then f is convex if and only if $\mathrm{dom} f$ is convex, and

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) \quad \text{for all } x, y \in \mathrm{dom} f$$



first-order approximation of f is global underestimator

Therefore, for a differentiable convex function, $\nabla f(x) = 0 \iff x$ minimizes f.

Key properties of convex functions (cont.)

Second-order condition:

If f is twice differentiable, then f is convex if and only if $\mathrm{dom} f$ is convex,

$$\nabla^2 f(x) \succeq 0 \quad \text{for all } x \in \mathrm{dom} f$$

Key properties of convex functions (cont.)

Second-order condition:

> If f is twice differentiable, then f is convex if and only if $\mathrm{dom} f$ is convex,
>
> $$\nabla^2 f(x) \succeq 0 \quad \text{for all } x \in \mathrm{dom} f$$

Example: **quadratic function** $f(x) = \frac{1}{2} x^T P x + q^T x + r,\ P \in \mathbb{S}^n$. Since

$$\nabla f(x) = Px + q, \quad \nabla^2 f(x) = P$$

Thus, $f(x)$ is convex if $P \succeq 0$.

Key properties of convex functions (cont.)

Second-order condition:

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$$\nabla f(x) = Px + q, \quad \nabla^2 f(x) = P$$

Thus, $f(x)$ is convex if $P \succeq 0$.

Example: **least-squares objective** $f(x) = \|Ax - b\|_2^2$ is always convex since

$$\nabla f(x) = 2A^T(Ax - b), \quad \nabla^2 f(x) = 2A^T A \succeq 0$$

Key properties of convex functions (cont.)

Example: **log-sum-exp function**: $f(x) = \log \sum_{k=1}^{n} \exp x_k$. Often called soft max, as it smoothly approximate $\max_{k=1,\ldots,n} x_k$.

It is convex, since:

$$\nabla_k f(x) = \frac{e^{x_k}}{\sum_{i=1}^{n} e^{x_i}}$$

$$\nabla^2 f(x) = \frac{1}{\mathbf{1}^T z}\mathrm{diag}(z) - \frac{1}{(\mathbf{1}^T z)^2} z z^T, \quad z_k = \exp x_k$$

and $\nabla^2 f(x) \succeq 0$ via $v^T \nabla^2 f(x) v \geq 0$ for any v by Cauchy-Schwarz inequality.

Key properties of convex functions (cont.)

α-sublevel set of $f : \mathbb{R}^n \to \mathbb{R}$: $C_\alpha = \{x \in \mathrm{dom} f : f(x) \leq \alpha\}$

> The sublevel sets of a convex function are convex.

Its converse is **false**.

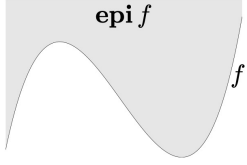
Key properties of convex functions (cont.)

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The sublevel sets of a convex function are convex.

Its converse is **false**.

Epigraph of $f : \mathbb{R}^n \to \mathbb{R}$: $\mathrm{epi} f = \{(x, t) \in \mathbb{R}^{n+1} : x \in \mathrm{dom} f, f(x) \leq t\}$



f is convex if and only if $\mathrm{epi} f$ is a convex set.

Operations preserving convexity

- Nonnegative linear combindation: $f_1, \ldots, f_m$ convex implies $a_1 f_1 + \cdots + a_m f_m$ convex for any $a_1, \ldots, a_m \geq 0$.

Operations preserving convexity

- Nonnegative linear combindation: $f_1, \ldots, f_m$ convex implies $a_1 f_1 + \cdots + a_m f_m$ convex for any $a_1, \ldots, a_m \geq 0$.
- composition with affine function: $f(Ax + b)$ is convex if f is convex.

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Example: (any) norm of affine function: $f(x) = \|Ax + b\|$

Example: log barrier for linear inequalities

$$f(x) = -\sum_{i=1}^{m} \log(b_i - a_i^T x), \quad \mathrm{dom} f = \{x : a_i^T x < b_i, i = 1, \ldots, m\}$$

Operations preserving convexity (cont.)

Pointwise supremum (pointwise maximization): if $f(x, y)$ is convex in x for each $y \in \mathcal{A}$, then

$$g(x) = \sup_{y \in \mathcal{A}} f(x, y)$$

is convex.

Operations preserving convexity (cont.)

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Examples:

- maximum distance to set C: $f(x) = \sup_{y \in C} \|x - y\|$
- support function of a set C: $S_C(x) = \sup_{y \in C} y^T x$ is convex
- maximum eigenvalue of symmetric matrix: for $X \in \mathbb{S}^n$, $\lambda_{\max}(X) = \sup_{\|u\|_2} u^T X u$

Operations preserving convexity (cont.)

Partial minimization: if $f(x, y)$ is convex in (x, y) and C is a convex set, then

$$g(x) = \inf_{y \in C} f(x, y)$$

is convex.

Operations preserving convexity (cont.)

Partial minimization: if $f(x, y)$ is convex in (x, y) and C is a convex set, then

$$g(x) = \inf_{y \in C} f(x, y)$$

is convex.

Example: minimum distance to set C:

$\operatorname{dist}(x, C) = \inf_{y \in C} \|x - y\|$ is convex if S is convex.

Operations preserving convexity (cont.)

Composition with scalar functions: suppose $f = h(g(x))$, where $g : \mathbb{R}^n \to \mathbb{R}$, $h : \mathbb{R} \to \mathbb{R}$. Then:

$$f \text{ is convex if } \begin{array}{l} g \text{ convex, } h \text{ convex, } \tilde{h} \text{ nondecreasing} \\ g \text{ concave, } h \text{ convex, } \tilde{h} \text{ nonincreasing} \end{array}$$

where $\tilde{h}$ is the extended-value function of h: $\tilde{h}(x) = h(x)$ if $x \in \mathrm{dom}(h)$, ∞ if $x \notin \mathrm{dom}(h)$.

How to remember these? For $n = 1$ and differentiable g, h:

$$f''(x) = h''(g(x))g'(x)^2 + h'(g(x))g''(x)$$

Operations preserving convexity (cont.)

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Examples:

- $\exp(g(x))$ is convex if g is convex
- $1/g(x)$ is convex if g is concave and positive

Operations preserving convexity (cont.)

Vector composition: suppose that

$$f(x) = h(g(x)) = h(g_1(x), \ldots, g_k(x))$$

where $g : \mathbb{R}^n \to \mathbb{R}^k$, $h : \mathbb{R}^k \to \mathbb{R}$, $f : \mathbb{R}^n \to \mathbb{R}$. Then:

f is convex if
- g convex, h convex, $\tilde{h}$ nondecreasing in each argument
- g concave, h convex, $\tilde{h}$ nonincreasing in each argument

Operations preserving convexity (cont.)

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- g convex, h convex, $\tilde{h}$ nondecreasing in each argument
- g concave, h convex, $\tilde{h}$ nonincreasing in each argument

Example: **softmax objective**: $\log \sum_{i=1}^m \exp g_i(x)$ is convex if g_i are convex. It smoothly approximates $\max_{i=1,\ldots,k} g_i(x)$.

Thank you!



References and further reading

- S. Boyd and L. Vandenberghe (2004), “Convex optimization”, Chapters 2 and 3.
- Dimitri P. Bertsekas (2009). "Convex optimization theory", Chapter 1.